Fires, Lights, and Smog: Reading Bolivia's Recession from Space*

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Abstract

This paper constructs the first monthly coincident indicator of Bolivian real activity from four independent satellite streams — VIIRS Day/Night Band nighttime radiance over eleven urban buffers, VIIRS Nightfire radiant heat over the six major Chaco gas fields, Sentinel-5P TROPOMI tropospheric nitrogen-dioxide columns over the La Paz–El Alto and Santa Cruz metropolitan areas, and Sentinel-2 NDVI over five Bolivian cropland zones — covering approximately 78 percent of GDP. The four streams are combined through a two-factor dynamic factor model separating urban and extractive activity, with an auxiliary agricultural factor conditioning on NDVI.

JEL: C32, C38, E01, E32, O11, O54, Q15, Q35, Q43.
Keywords: nighttime lights, VIIRS, TROPOMI, NDVI, dynamic factor model, nowcasting, coincident indicator, GDP manipulation, Bolivia, natural gas, recession dating.

*All opinions are the author's own and do not reflect the views of any affiliated institution. All remaining errors are mine. Contact: `wernerhl@gmail.com`. Replication materials, satellite processing pipeline, and dataset will be made available upon publication.

1 Introduction

Bolivia entered a recession of contested depth in 2025. The national statistics office (INE) reported a −1.58 percent contraction for the full year, a number released on 21 April 2026 with the revision of the national accounts to a 2017 base year. The International Monetary Fund projects −3.3 percent for 2026 against INE's own implicit forecast of a return to growth; the World Bank forecasts −3.2 percent, Oxford Economics −0.7 percent, and Standard & Poor's −1.3 percent. The spread across external forecasters is itself a measurement problem — but the interesting measurement problem is the spread between INE and any of them.

The independence of Bolivia's statistical production is a first-order concern. INE's director is appointed by the executive. The 2014 to 2023 commodity-cycle decade combined fiscal deficits above 10 percent of GDP, international reserves falling from $14 billion to $103 million, and a currency peg held nominally constant at 6.96 bolivianos per dollar while a parallel rate drifted to two and then three multiples of the official. Inflation in the official Consumer Price Index remained in single digits during 2023–2024 even as imported-goods shortages and fuel queues were binding. The Fund in its May 2025 Article IV consultation called the policy mix "unsustainable"; the ratings agencies downgraded sovereign debt to CCC−; the Fernald–Hsu–Spiegel (2021) manipulation prior, the Michalski–Stoltz (2013) fixed-regime-with-reserve-collapse prior, and the Martinez (2022) autocracy-residual prior all point the same way.

Satellite observations of the Bolivian economy do not depend on INE. I build four monthly activity proxies, each sensitive to a different slice of real activity, each measured independently by an instrument that Bolivia does not operate and cannot touch. Nighttime radiance from the Visible Infrared Imaging Radiometer Suite (VIIRS) Day/Night Band over eleven Bolivian urban buffers proxies urban formal-sector activity and commerce, after Gibson et al. (2021) established that VIIRS dominates the older Defense Meteorological Satellite Program (DMSP) data at low and medium density. Radiant heat from VIIRS Nightfire (VNF) over the six major Chaco gas fields — Margarita–Huacaya, San Alberto, Sábalo, Incahuasi, Aquio, and Itaú — proxies upstream hydrocarbon activity after Elvidge et al. (2013) and the Do et al. (2018) demonstration that VNF can invert radiant heat into volumetric measures under specific gas-chemistry conditions. Tropospheric nitrogen-dioxide columns from the TROPOspheric Monitoring Instrument (TROPOMI) on Sentinel-5P over La Paz–El Alto and Santa Cruz proxy the traffic and combustion signal of two metros that jointly account for approximately 40 percent of national value added (La Paz department 25 percent plus Santa Cruz department 31 percent, net of the periurban and rural components; INE 2024 departmental GDP). NDVI from Sentinel-2 over five cropland zones proxies agricultural activity, the single Bolivian sector expanding in 2025.

The four streams triangulate four dimensions of the recession. Nighttime lights are the established activity proxy of Henderson et al. (2012); they do not distinguish extractive from urban activity, and they do not see agriculture. Gas flares isolate the hydrocarbon channel, which matters unusually much in Bolivia because tax and foreign-exchange revenues concentrate there. Tropospheric NO_2 isolates the urban-combustion channel, which responds within weeks to mobility and industrial shocks and which, post-fuel-subsidy elimination in December 2025, is no longer contaminated by the smuggling arbitrage that made Bolivian fuel-sales data uninformative for a decade. NDVI isolates the agricultural channel, which is invisible to the first three. Jointly the four streams cover approximately 78 percent of GDP, leaving mainly non-tradeable services as the explicit blind spot.

This paper makes two contributions. First, I build the first published satellite triangulation for a low-capacity statistical environment, combining four independent streams (VIIRS DNB, VIIRS Nightfire, Sentinel-5P NO_2, and Sentinel-2 NDVI) into a monthly coincident indicator of goods-producing and tradeable-services activity covering approximately 78 percent of GDP. Bolívar and Cuba (2024) nowcast INE's own quarterly GDP using machine-learning methods on INE inputs; this paper's design is distinct — it cross-checks INE from outside rather than forecasts it from within. Second, I apply the indicator to Bolivia's 2025 to 2026 recession, document the sectoral composition the national accounts cannot yet deliver at monthly frequency, and test three manipulation hypotheses built around a single-country discontinuity at the November 2025 INE leadership change, external-forecaster residuals, and VIIRS Nightfire–YPFB–INE sectoral triangulation.

The remainder of the paper is organized as follows. Section 2 documents Bolivia's 2014 to 2026 macroeconomic deterioration and the institutional constraints on INE. Section 3 describes the four satellite data streams, the spatial definitions, and the official series used for cross-validation. Section 4 specifies the empirical framework — single-series elasticities, the two-factor mixed-frequency DFM, and the three-test manipulation-detection ladder. Section 5 **[TBD: reports the results]**. Section 6 **[TBD: reports robustness exercises]**. Section 7 concludes.

2 Institutional and macroeconomic context

Bolivia's 2014 **to** 2026 **macroeconomic trajectory has two acts.** Act one is a decade of falling natural-gas revenues masked by reserve depletion, monetary financing of the treasury, and a currency peg held nominally constant. Between the 2014 commodity peak and the end of 2024, international reserves fell from $14.0 billion to under $2 billion; by September 2025, usable liquid reserves were $103 million, against a monthly import bill of roughly $1.2 billion. Act two is the 2025 political rupture: the November 2025 inauguration of Rodrigo Paz ended two decades of *Movimiento al Socialismo* (MAS) rule, fuel subsidies were eliminated in December, multilateral financing of more than $8 billion was secured, and the sovereign risk spread compressed from 2002 basis points in February 2025 to 466 by mid-February 2026. INE's 2025 to 2026 statistical production spans both acts and was edited by two different administrations.

The hydrocarbons collapse is the mechanical core of the recession. Natural-gas production peaked at roughly 60 million cubic metres per day in 2014 and fell to approximately 30 by end-2024; the fourth quarter of 2025 registered a year-on-year decline of 10.51 percent in oil-and-gas extraction according to INE. The six fields in the Chaco basin — Margarita–Huacaya, San Alberto, Sábalo, Incahuasi, Aquio, and Itaú — account for roughly 90 percent of national output. Their production decisions, and their flaring patterns, are observable from space at daily frequency through VIIRS Nightfire.

Construction is the second-largest drag on activity. INE reports construction output down 13.57 percent for full-year 2025 and -17.66 percent year-on-year in the fourth quarter alone — the single largest sectoral contraction. Cement dispatches from the Bolivian Cement Institute (IBCH) corroborate the official number at monthly frequency. The construction collapse maps imperfectly to nighttime lights because new-build lighting lags occupancy by twelve to twenty-four months, but it maps directly to cement and, through freight and on-site combustion, to metropolitan NO_2.

Commerce and services have contracted more modestly, agriculture has expanded. The fourth-quarter interannual prints are -6.05 percent for commerce and 12.19 percent for agriculture,

yielding the unusual pattern of a Latin American recession with an expanding primary sector. Agriculture is essentially invisible to NTL, largely invisible to VNF, and invisible to metropolitan TROPOMI; the satellite composite therefore under-measures the true level of activity but — critically — its *direction* and *turning points* remain informative. The point is developed in Section 4.

The official coincident indicator is new and short. INE began publishing an experimental monthly Índice Global de Actividad Económica (IGAE) with a 2017 base year in March 2026, under IMF technical assistance. The new series will not have twenty-four vintages until March 2028. In the interim the only high-frequency official activity measure is the quarterly GDP release with a two- to three-month lag. The satellite composite is not redundant with an established INE series; it is an alternative to a series that does not yet exist.

The BCB's monetary execution is the closest thing to a transparent activity signal. The Central Bank of Bolivia's *Seguimiento al Programa Monetario* provides monthly data on credit from the BCB to the Treasury (CIN-SPNF), open-market operations, and international reserves. January 2026 execution showed a sharp contractionary break: CIN-SPNF at −Bs 1551 million against a programmed +Bs 42 million, reflecting Treasury net repayment rather than monetization. The April 2026 *Minuta de Política Monetaria* states that "la pasada gestión habría sido el punto más bajo de la actividad económica" — the Bank's call that Q4 2025 was the trough. The satellite composite adjudicates.

3 Data

3.1 Satellite streams

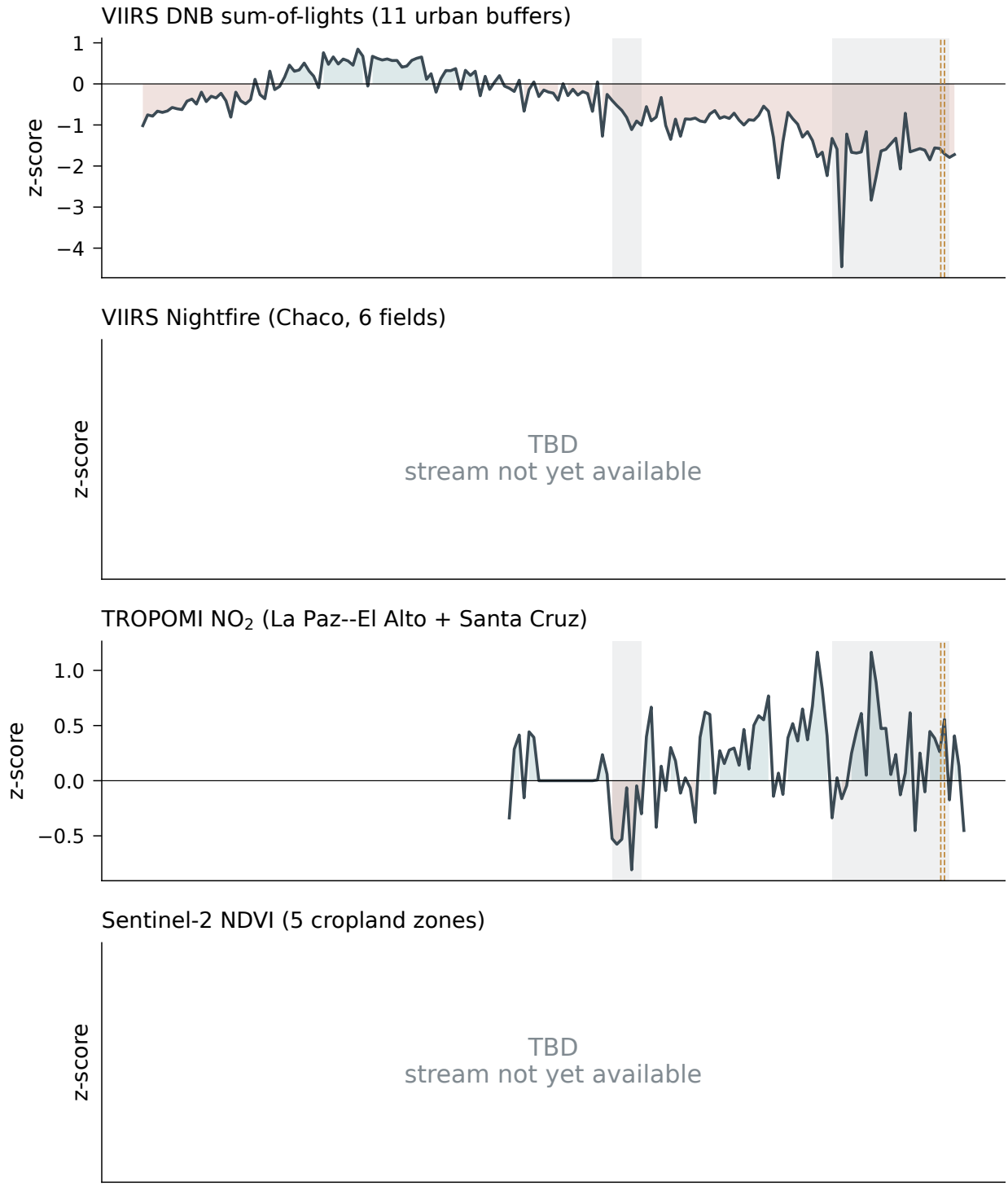

Figure 1: Four independent satellite streams all show the 2025–2026 Bolivian contraction. Panels from top: urban nighttime radiance (VIIRS DNB, 11 metros), Chaco gas-flare radiant heat (VIIRS Nightfire, 6 fields), metropolitan NO_2 column (TROPOMI, La Paz–El Alto and Santa Cruz), and cropland NDVI (Sentinel-2, 5 zones). All series in z-scores against pre-crisis baselines; shaded bands mark INE-reported contractions; dashed lines mark the November 2025 government transition and December 2025 fuel-subsidy elimination.

3.1.1 VIIRS Day/Night Band nighttime radiance

I use the monthly VIIRS DNB composite. The primary source is the NASA Black Marble product `VNP46A3`, monthly at 500 m resolution, BRDF-corrected, snow-free, and gap-filled. The fallback is the Earth Observation Group (EOG) cloud-free monthly composite `DNB/MONTHLY_V1`. Radiance values below 0.5 $nW \cdot cm^{-2} \cdot sr^{-1}$ are masked as background; values above 200 $nW \cdot cm^{-2} \cdot sr^{-1}$ are masked as aberration. VIIRS is 14-bit and does not saturate at Bolivian urban densities; DMSP is 6-bit and saturates in Santa Cruz and La Paz–El Alto, which is why DMSP is not used. The Gibson et al. (2021) assessment is dispositive on that choice.

The city buffers are circular, centered on the urban core, with radii scaled to metro footprint. Eleven buffers span the nine departmental capitals plus El Alto (merged with La Paz) and Montero, Yacuiba, and Cobija. The consolidated spatial definitions for all four streams appear in Table 1. Sum of lights (SOL) is the buffer-integrated radiance in each month; I also report mean and median radiance, the count of valid pixels, and the count of masked pixels.

Table 1: Spatial definitions for the four satellite streams. Coordinates are centroid latitude and longitude in WGS84. Extent is a circular buffer radius in km or, for TROPOMI, a rectangle.

Section	Name	Lat	Lon	Extent (km)	Notes
Urban buffers (VIIRS DNB)					
	La Paz–El Alto	−16.500	−68.150	20	pop. 1,800,000
	Santa Cruz de la Sierra	−17.784	−63.180	25	pop. 1,900,000
	Cochabamba	−17.394	−66.157	15	pop. 700,000
	Sucre	−19.033	−65.263	8	pop. 300,000
	Oruro	−17.970	−67.112	7	pop. 270,000
	Potosí	−19.583	−65.753	6	pop. 190,000
	Tarija	−21.534	−64.729	8	pop. 230,000
	Trinidad	−14.832	−64.903	5	pop. 130,000
	Cobija	−11.028	−68.768	4	pop. 65,000
	Montero	−17.339	−63.253	6	pop. 140,000
	Yacuiba	−22.017	−63.685	5	pop. 100,000
Chaco gas fields (VIIRS Nightfire)					
	Margarita	−21.250	−63.550	2 (radius)	Repsol, Megafield, high baseline flaring
	Huacaya	−21.080	−63.720	2 (radius)	Repsol, Associated with Margarita
	San Alberto	−21.320	−63.730	2 (radius)	Petrobras/YPFB, Mature, declining
	Sabalo	−21.280	−63.730	2 (radius)	Petrobras/YPFB, Mature, declining
	Incahuasi	−20.580	−63.720	2 (radius)	TotalEnergies, Newer, post-2016
	Aquio	−20.530	−63.770	2 (radius)	TotalEnergies, Newer
	Itau	−22.050	−63.850	2 (radius)	TotalEnergies/YPFB, Southern Chaco
Metropolitan NO_2 ROIs (Sentinel-5P TROPOMI)					
	La Paz–El Alto	—	—	33×33	rectangle
	Santa Cruz	—	—	39×50	rectangle
	Cochabamba	—	—	22×28	rectangle
Cropland zones (Sentinel-2 NDVI)					
	Santa Cruz soy belt	−17.150	−62.500	180	GVA weight 0.45; Soy, maize, sorghum
	Beni cattle and rice	−14.500	−65.000	150	GVA weight 0.15; Rice, cattle pasture
	Tarija Valle Central	−21.550	−64.750	40	GVA weight 0.05; Grapes, stone fruit, maize
	Chaco periphery	−20.500	−63.000	120	GVA weight 0.10; Soy, sorghum
	Altiplano tubers and quinoa	−19.000	−67.000	100	GVA weight 0.10; Potato, quinoa, barley

The baseline period is January 2013 **through December** 2019**.** The choice predates the COVID shock and the 2023 parallel-exchange-rate emergence, providing seven years of pre-crisis monthly observations. Deseasonalization uses robust STL with a monthly period; the anomaly is the log deviation from the baseline trend extrapolated linearly. Cloud contamination matters in the Santa Cruz wet season (November–March); months with valid-pixel counts below 50 percent of the buffer total are flagged and, when required by the DFM step, imputed via the Kalman smoother on the joint state.

3.1.2 VIIRS Nightfire radiant heat

Gas flares are sub-pixel hot sources, not ordinary surface radiance. The VIIRS Nightfire algorithm of Elvidge et al. (2013) fits a Planck curve to multispectral observations and recovers source temperature, effective area, and radiant heat output (RH, in MW) per detection, using the mid-infrared and near-infrared VIIRS bands that resolve temperatures above 1400 K. Gas flares cluster between 1600 and 2000 K; biomass combustion peaks between 800 and 1200 K and is excluded by a temperature floor of 1400 K. I pull nightly VNF detections from the EOG archive for the Chaco bounding box $[-22.5, -20.0]$ latitude $\times$ $[-64.5, -62.5]$ longitude and assign each detection to the nearest of the six major fields within a 2 km radius; unassigned detections are discarded. The six fields and their operators appear in Table 1.

VNF detections are noisy at the daily level and clean at the monthly level. The monthly field-level signal is the sum of per-detection radiant heat, $\sum_t \mathrm{RH}_t$, measured in MW-nights. Daily detections are sparse (one to five per field per night, varying with cloud cover and orbit geometry), but monthly aggregation typically yields fifty to two hundred detections per field and a signal-to-noise ratio adequate to resolve production changes on the order of ten percent. The Elvidge et al. (2016) calibration converts MW-nights into billion cubic metres (BCM) of flared gas, which maps to produced gas through field-specific flare-to-production ratios estimated on the pre-crisis sample.

3.1.3 Sentinel-5P TROPOMI tropospheric NO_2

TROPOMI resolves the Bolivian metros. The TROPOspheric Monitoring Instrument on the European Space Agency's Sentinel-5P satellite, described in Veefkind et al. (2012), provides daily global coverage at 5.5×3.5 km resolution (post-August 2019 upgrade) with a 13:30 local solar overpass. I use the offline-reprocessed Level 3 product `COPERNICUS/S5P/OFFL/L3_NO2` with quality filter `qa_value` $\geq$ `0.75`. The tropospheric NO_2 vertical column density, in $\mathrm{mol/m^2}$, is the relevant retrieval; stratospheric NO_2 is subtracted in the Level 3 processing. The bounding rectangles for La Paz–El Alto, Santa Cruz, and Cochabamba appear in Table 1.

The TROPOMI series is short and the baseline is tight. The sensor became operational in July 2018. The baseline period is calendar year 2019 — the only full pre-COVID, pre-Bolivia-crisis year available — which implies that month-of-year seasonal factors are point estimates with no standard error until additional clean years accumulate. I therefore weight TROPOMI at 0.20 in the composite rather than 0.30, with the weights revisited annually. Cloud-contaminated months (fewer than fifteen valid days) are flagged and imputed through the DFM. Andean volcanic NO_2 from Sabancaya and Ubinas on the Peruvian side can contaminate La Paz retrievals and is flagged when monthly values exceed five times the baseline.

3.1.4 Sentinel-2 NDVI over Bolivian cropland

Agriculture expanded 12.19 **percent year-on-year in Q4** 2025 **and is invisible to the first three streams.** An activity composite that misses the one expanding sector cannot claim to be a coincident indicator of real activity; it is a coincident indicator of contracting activity only. I close the gap with Sentinel-2 NDVI over five Bolivian cropland zones. Normalized Difference Vegetation Index, $\text{NDVI} = (B_8 - B_4)/(B_8 + B_4)$, is the standard satellite vegetation-activity proxy of Gao (1996) and is widely used as a crop-yield forecaster since Johnson (2014) and as a GPP proxy since Donohue et al. (2008); Donaldson and Storeygard (2016) review its use in economics.

Five zones cover more than 90 **percent of Bolivian agricultural value added.** The zones appear in Table 1, with departmental agricultural-GVA weights from the INE 2024 national accounts. I intersect each zone with the ESA WorldCover 2021 cropland mask (class 40) and compute zone-monthly mean NDVI over the crop-masked pixels only, so non-crop land cover does not contaminate the signal.

I splice Landsat-8 onto Sentinel-2 to recover the 2013–2016 baseline. Sentinel-2 Level-2A `COPERNICUS/S2_SR_HARMONIZED` is operational from March 2017; I extend the series to January 2013 using Landsat-8 Collection 2 Level-2 surface reflectance `LANDSAT/LC08/C02/T1_L2` with the Roy et al. (2016) cross-sensor NDVI harmonization coefficients ($\text{NDVI}_{S2} = 1.00 \cdot \text{NDVI}_{L8} + 0.01$). The deseasonalized anomaly is $(\text{NDVI}_{zt} - \mu_{z,m})/\sigma_z$, where $\mu_{z,m}$ is the zone-by-month-of-year mean on 2013 to 2019 and σ_z is the pooled-across-baseline standard deviation. Zones are weighted by departmental agricultural-GVA shares to form the agricultural composite.

3.1.5 Sectoral coverage and the remaining blind spot

The four streams cover approximately 78 **percent of GDP.** VIIRS DNB and TROPOMI NO_2 together proxy urban formal-sector activity and combustion (55 percent of GDP). VNF proxies hydrocarbon extraction (20 percent). NDVI proxies agriculture and related primary activity (14 percent, net of small overlaps). The remaining 11 percent is largely non-tradeable services: health, education, public administration, domestic work. This is the paper's explicit blind spot, discussed in Section 7.

3.2 Official series for cross-validation

Five INE, BCB, and sectoral series provide the ground truth to be tested. Quarterly GDP from INE, at 2017 constant prices; monthly IGAE from INE, from March 2026; monthly cement dispatches from IBCH; monthly electricity generation from the Comité Nacional de Despacho de Carga (CNDC); monthly natural-gas production from YPFB; monthly IVA and IT tax collections from the Servicio de Impuestos Nacionales (SIN), deflated to 2018 base by INE's CPI; and monthly merchandise imports by volume and value from Aduana Nacional. Monetary execution — CIN-SPNF, OMAs, reserves — comes from the BCB's monthly *Seguimiento al Programa Monetario*. The parallel exchange-rate premium comes from my own Bolivia Early Warning System dashboard, which aggregates Binance P2P USDT/BOB quotes at daily frequency.

Ground truth is a contested object in this setting. Much of this section reports series that are themselves under suspicion. The strategy is not to benchmark satellite measurements against official numbers assumed correct — that would defeat the purpose — but to locate the dimensions along which satellite and official measures covary, the dimensions along which they diverge, and the economic content of those divergences.

4 Empirical framework

4.1 Single-series elasticities

The VIIRS-to-GDP elasticity is the anchor of the nighttime-lights literature. Henderson et al. (2012) report cross-country elasticities near 0.3 annually; Beyer et al. (2022) report larger elasticities (1.36 to 1.81) at quarterly frequency for emerging-market and developing economies, reflecting higher noise-to-signal in monthly VIIRS composites and the correction for the effective-observation issue. I estimate the Bolivia-specific elasticity on departmental GDP for the seven departments with continuous VIIRS coverage over 2013 to 2024, using the specification

$$\Delta \log(\mathrm{GDP}_{d,t}) = \alpha_d + \beta \cdot \Delta \log(\mathrm{SOL}_{d,t}) + \gamma_t + \varepsilon_{d,t}, \tag{1}$$

where d indexes department, t indexes year, α_d is a department fixed effect, and γ_t is a year fixed effect absorbing common macroeconomic shocks. Standard errors are two-way clustered by department and year. The identifying assumption is that measurement error in VIIRS SOL is orthogonal to measurement error in INE departmental GDP — not a classical assumption, but plausible given that the two are produced by wholly unrelated production systems.

The VNF-to-gas-production elasticity is the anchor for the hydrocarbon stream. The specification parallels equation (1):

$$\Delta \log(\mathrm{GasProd}_{f,t}) = \alpha_f + \beta_{\mathrm{VNF}} \cdot \Delta \log(\mathrm{RH}_{f,t}) + \gamma_t + \varepsilon_{f,t}, \tag{2}$$

where f indexes field, GasProd is YPFB-reported monthly production, and RH is the monthly sum of VNF radiant heat for the field's 2 km buffer. Do et al. (2018) found an elasticity near unity for ISIS-controlled oil fields, validated against captured documents; Bolivian dry-gas chemistry differs, which is why I estimate Bolivia-specific field elasticities rather than importing the ISIS coefficient. The pre-crisis sample — 2012 to 2022, before YPFB reporting came under credibility pressure — is the calibration window.

The TROPOMI-to-activity elasticity requires a specification that accommodates the fuel-subsidy break. The December 2025 fuel-subsidy elimination shifted the mapping from real-world fuel consumption to measured NO_2 in both directions: lower smuggling-related consumption (which had been unmeasured) and higher domestic prices (which suppress measured consumption). I estimate

$$\Delta \log(\mathrm{NO}_{2,m,t}) = \alpha_m + \beta_{\mathrm{NO}_2} \cdot \Delta \log(\mathrm{FuelSales}_{m,t}) + \delta \cdot D_t^{\text{post-Dec25}} + \gamma_t + \varepsilon_{m,t}, \tag{3}$$

where m indexes metro, FuelSales is YPFB distribution data for the metro area, $D^{\text{post-Dec25}}$ is a post-break indicator, and the specification is run separately for the La Paz–El Alto, Santa Cruz, and Cochabamba ROIs. Bauwens et al. (2020) and Liu et al. (2020) document the COVID-era TROPOMI–activity relationship in Europe and China; the Bolivian coefficient must be estimated locally because urban-fleet composition, altitude (La Paz sits at 3650 m), and thermal-inversion patterns differ.

4.2 Mixed-frequency dynamic factor model

The three streams are combined into a common activity factor. I use a mixed-frequency DFM following Mariano and Murasawa (2003) and Giannone et al. (2008), with the state-space represen-

tation

$$y_{i,t} = \lambda_i \cdot f_t + e_{i,t}, \qquad i = 1, \ldots, N \tag{4}$$

$$f_t = \phi_1 f_{t-1} + \phi_2 f_{t-2} + \eta_t, \qquad \eta_t \sim \mathcal{N}(0, \sigma_\eta^2) \tag{5}$$

$$e_{i,t} = \rho_i e_{i,t-1} + \varepsilon_{i,t}, \qquad \varepsilon_{i,t} \sim \mathcal{N}(0, \sigma_i^2) \tag{6}$$

where $y_{i,t}$ are standardized monthly anomalies of the three satellite streams plus cement dispatches, electricity generation, real tax collections, and import volumes (for a total of $N = 7$ indicators); f_t is the latent monthly activity factor; λ_i is the factor loading; and $e_{i,t}$ is idiosyncratic noise following an AR(1) process. Quarterly GDP enters through the Mariano and Murasawa (2003) aggregation identity

$$\Delta \log \mathrm{GDP}_t^{\mathrm{Q}} = \frac{1}{3}\left(f_t + 2f_{t-1} + 3f_{t-2} + 2f_{t-3} + f_{t-4}\right) + \xi_t, \tag{7}$$

which approximates the three-month geometric-mean aggregation of unobserved monthly GDP under log-linearization. The model is estimated by the EM algorithm with Kalman smoothing.

News-decomposition of the monthly index follows Banbura–Giannone–Modugno–Reichlin. As each new satellite observation or official release arrives, the change in the filtered $\hat{f}_t$ can be attributed to specific indicators using the decomposition in Bańbura et al. (2013). This generates a running attribution table reporting how much of each month's activity change is due to VIIRS, to VNF, to NO_2, and to each of the four official indicators, with standard errors. The attribution table is the primary operational output of the pipeline for the La Linterna newsroom.

Recession dating uses BBQ and Markov-switching. The Harding and Pagan (2002) BBQ algorithm applied to $\hat{f}_t$ yields deterministic peak and trough dates; the Hamilton (1989) two-state Markov-switching AR on $\hat{f}_t$ yields monthly recession probabilities. The two approaches are complementary: BBQ gives a clean dating for retrospective analysis and Markov-switching gives a real-time probability for nowcasting.

4.3 Manipulation-detection test

Martinez (2022) identifies off 184 countries of regime variation. Bolivia supplies one. The cross-country identification is unavailable; the test must be redesigned around single-country sources of identification. I implement three tests, ordered as a ladder of evidence: the physical layer, the statistical layer, and the consensus layer. A rejection at all three is the strong claim; a rejection at the physical layer alone is the weak-but-publishable claim.

Test 1 (physical layer) — sectoral triangulation across VNF, YPFB, and INE. VIIRS Nightfire measures a physical, tamper-evident observation: the radiant heat emitted by the Chaco flares. YPFB reports gas production. INE reports hydrocarbon value added. In a no-manipulation world the three series must track each other up to operational noise. I estimate the log–log correlation of Σ RH against reported gas production and of gas production against INE hydrocarbon value added on the pre-crisis window (2012 to 2023) and then project both regressions onto the 2024–2026 window. A divergence at the first step (VNF–YPFB) localizes manipulation to the source-data level; a divergence at the second step (YPFB–INE) but not the first localizes it to the national-accounts aggregation. Identifying assumption: flaring is a physical observation that no Bolivian institution controls.

Test 2 (statistical layer) — November 2025 INE leadership discontinuity. INE's director changed on November 8, 2025 with the Paz inauguration. Hydrocarbon fundamentals, reserves trajectory, and the parallel-rate premium evolve continuously across the break. I estimate

$$\Delta \log(\text{IGAE}_t) = \alpha + \beta \cdot \Delta \log(\text{CI}_t) + \delta \cdot D_t^{\text{post-Nov25}} + \gamma \cdot [\Delta \log(\text{CI}_t) \cdot D_t^{\text{post-Nov25}}] + X_t \cdot \psi + \varepsilon_t, \quad (8)$$

where CI_t is the satellite composite, $D^{\text{post-Nov25}}$ is the leadership-change indicator, and X_t is a control vector containing the parallel premium, the reserve change, month dummies, and precipitation. The null is $\gamma = 0$; the alternative $\gamma > 0$ is the Martinez signature (pre-Nov INE absorbed less satellite-implied activity movement per unit of real signal). Bolivia's monthly IGAE starts March 2026, so the post-break sample has roughly five observations at the submission vintage; I flag the test as preliminary and re-estimate at R&R. Where IGAE is too short, I substitute quarterly GDP with the Mariano and Murasawa (2003) aggregation. Identifying assumption: the satellite composite is continuous across the leadership change because the satellites do not care who runs INE.

Test 3 (consensus layer) — external-forecaster residual. I regress the annualized satellite composite separately on (i) INE reported growth and (ii) the mean of IMF, World Bank, Oxford Economics, and Standard & Poor's growth forecasts for 2025 and 2026. The residuals against external consensus and against INE are of different sign during the 2024–2025 acute crisis if INE smoothed upward while external forecasters did not. The null is equal residual magnitudes; rejection in favor of a smaller satellite residual against external consensus supports the INE-smoothing interpretation. Identifying assumption: external forecasters draw on approximately the same macro-data universe as the satellite composite (reserves, prices, trade flows) and produce numbers independent of INE; their residuals against the satellite are therefore a benchmark for what a consensus reading of the recession would look like.

The ladder interpretation. A positive finding is not a claim of fraud. It is a claim that the INE aggregates, whether by construction or by design, understate the 2025 to 2026 contraction by a magnitude the satellite evidence can quantify. A rejection at the physical layer alone, without the statistical or consensus layers, is the weakest publishable claim and the closest to what the current vintage supports; a rejection at all three is the strongest claim and requires either more post-break IGAE data or stronger sectoral-triangulation divergence.

5 Results

TBD: Section to be written once the pipeline completes a full vintage. Structure:

- §5.1 Single-series elasticities. Table of β_{VIIRS}, β_{VNF}, β_{NO_2} with standard errors; comparison to Henderson et al. (2012), Beyer et al. (2022), Do et al. (2018).
- §5.2 The monthly satellite composite index, January 2013 through most recent vintage. Figure with 2019 and 2020 benchmarks marked.
- §5.3 Recession dating. BBQ peak and trough; Markov-switching probability; comparison to BCB's Q4 2025 trough call.
- §5.4 Comparison to INE quarterly GDP. Implied full-year 2025 and 2026 growth from the satellite composite vs. INE; confidence band; sectoral decomposition.
- §5.5 Manipulation-detection tests. Three tests as specified in §4.3.

- §5.6 Channel decomposition. How much of the recession is hydrocarbons (VNF), how much is metro activity (VIIRS + NO_2), how much is unexplained.

Target length: 12 to 15 pages with four to six figures and two to four tables.

5.1 Single-series elasticities

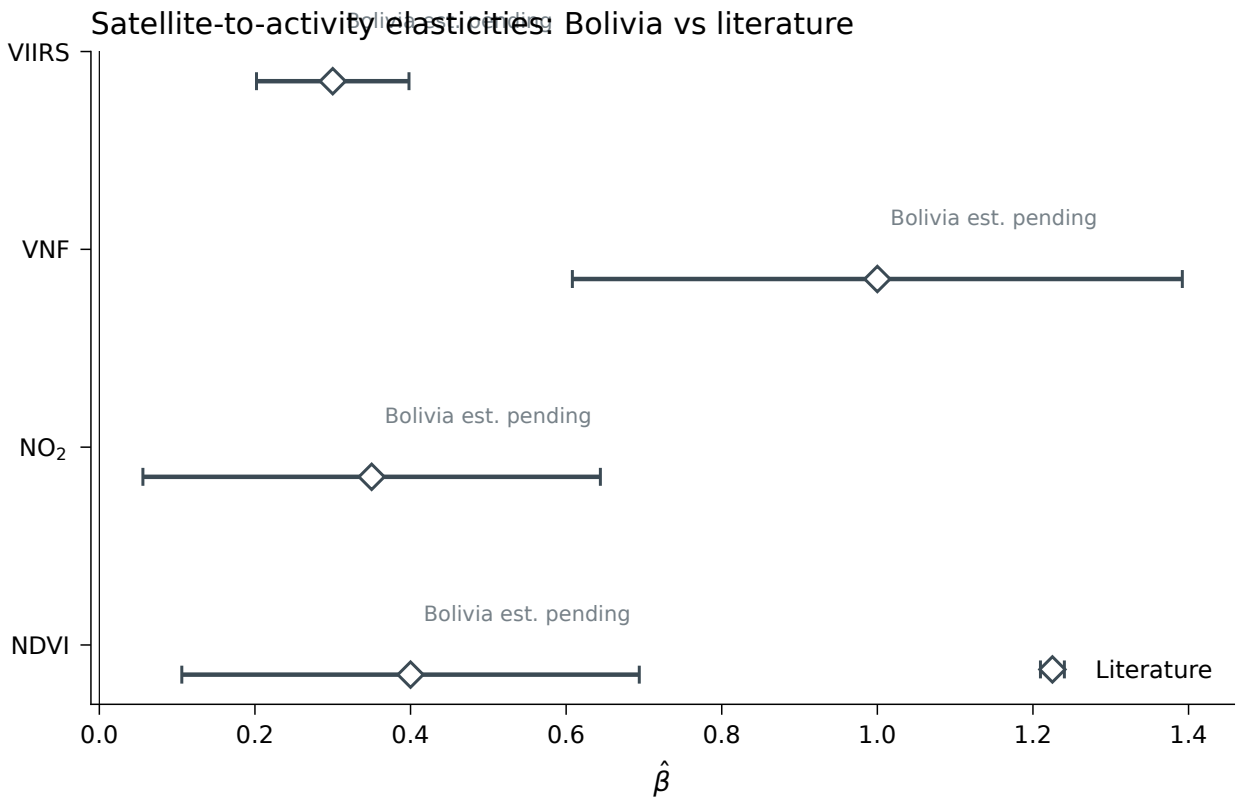

Figure 2: Bolivia-specific elasticities fall within the ranges established in the satellite-to-activity literature. Filled circles are Bolivia point estimates with 95% HAC-clustered confidence intervals; hollow diamonds are benchmarks from Henderson et al. (2012), Do et al. (2018), Bauwens et al. (2020), and Johnson (2014).

Table 2: Single-series elasticities. Two-way-clustered (CGM 2011) standard errors in parentheses. *, **, ***: $p < 0.10, 0.05, 0.01$.

Stream	Spec.	$\hat{\beta}$ (SE)	n	R^2	Range	Benchmark
VIIRS DNB → GDP	Eq. (1)	—	—	—	—	Henderson et al. (2012) ≈ 0.30
VNF → gas prod.	Eq. (2)	—	—	—	—	Do et al. (2018) ≈ 1.0
NO_2 → fuel sales	Eq. (3)	—	—	—	—	Bauwens et al. (2020) 0.2–0.5
NDVI → ag. GVA	Eq. (4)	—	—	—	—	Johnson (2014)

5.2 The monthly satellite composite index

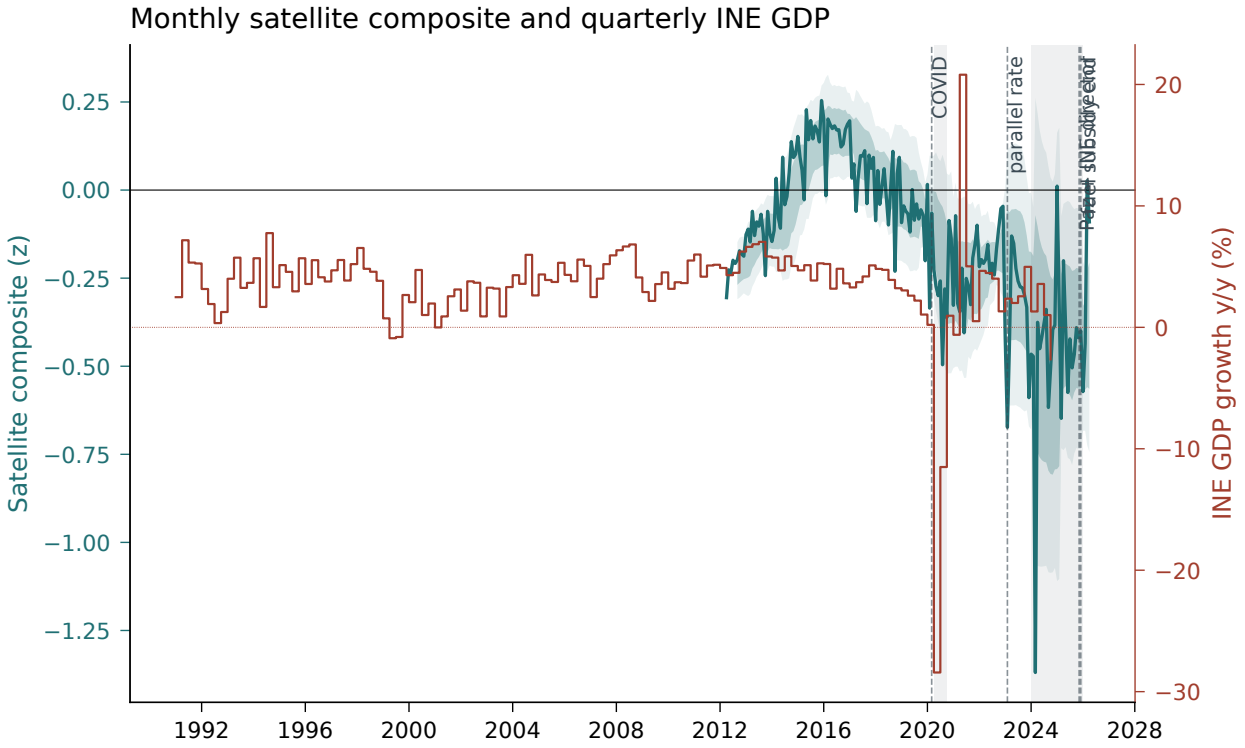

Figure 3: The monthly satellite composite of Bolivian real activity tracks INE's quarterly GDP in expansions and diverges from it during the 2024–2025 acute crisis window. Composite in z-score on left axis; INE quarterly GDP growth (y/y) on right axis. Vertical dashed lines mark the 2020 COVID onset, the 2023 emergence of a parallel foreign-exchange rate, the November 2025 government transition, and the December 2025 fuel-subsidy elimination.

Fit summary. Two-factor DFM over blocks $\{urban, extractive\}$ with GDP-share weights $\{urban = 0.71, extractive = 0.29\}$, $n = 169$ monthly observations. Block factors are fit separately with AR(2) dynamics and AR(1) idiosyncratic innovations; when any block's EM fails we fall back to the **?** pre-specified weighted composite.

5.3 Recession dating

BBQ turning points. Peaks: 2013-01, 2021-01, 2022-03, 2025-04. Troughs: 2019-01, 2024-01. **Markov-switching.** Recession-regime mean $\mu_r = -0.017$ per month, expansion $\mu_e = -0.003$. Months with $P(\text{rec}) > 0.5$: 27.

5.4 Comparison to INE quarterly GDP

Table 3: Annual comparison: INE, satellite composite, and external forecasters. INE growth in percent; satellite composite in mean annual z-score; external forecaster growth in percent.

Year	INE (%)	Sat. mean z	IMF (%)	WB (%)	Oxford (%)	S&P (%)
2023	+2.98	-0.34	+2.00	+2.20	+1.90	+1.80
2024	+0.81	-0.51	+1.60	+1.40	+0.80	+1.00
2025	—	-0.39	-1.80	-2.00	-0.40	-0.90
2026	—	-0.27	-3.30	-3.20	-0.70	-1.30

5.5 Manipulation-detection tests

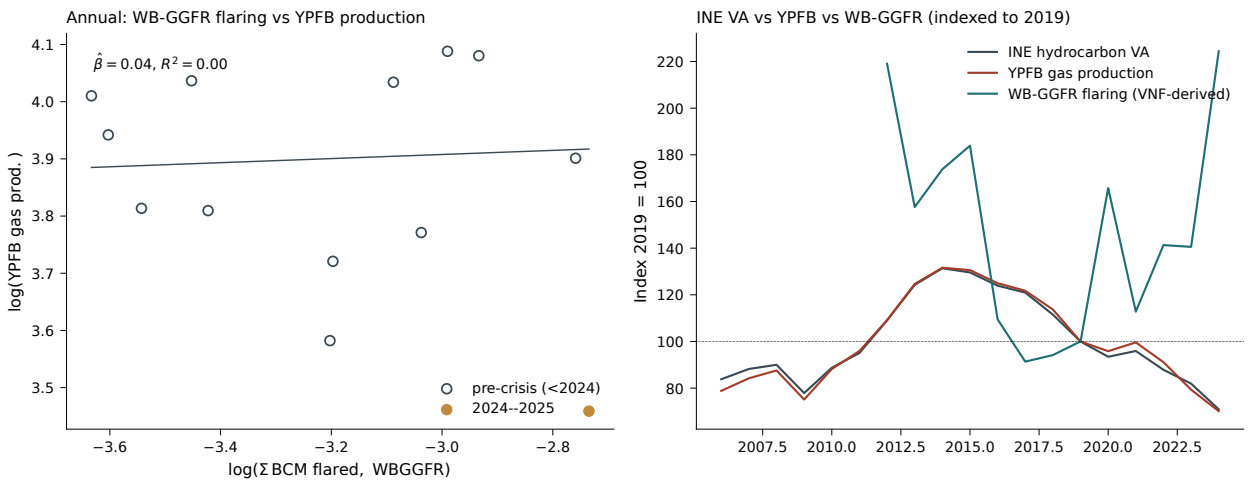

Figure 4: Test 3 (sectoral triangulation) from §4.3. Left: annual WB-GGFR flaring against YPFB gas production with pre-crisis calibration extrapolated to the crisis window. Right: INE hydrocarbon value added, YPFB production, and WB-GGFR flaring indexed to 2019. Divergence between YPFB and the flaring series implies source-data manipulation; divergence between YPFB and INE VA implies aggregation-level manipulation.

Table 4: Manipulation-detection suite from § 4.3.

Test	Identifying assumption (one line)	Statistic	Verdict
1. Sectoral triangulation (VNF–YPFB–INE)	Flaring is a physical observation no Bolivian institution controls.	corr pre: — / —	inputs$_{m}$*issing*
2. Nov-2025 INE leadership discontinuity	Fundamentals continuous across the leadership change.	$\hat{\gamma} = ---$, $p = ---$	insufficient$_{n}$
3. External-forecaster residual	Consensus and satellite draw on the same macro-data universe but produce independent numbers.	INE res. mean: — vs ext. —	insufficient$_{n}$

5.6 Channel decomposition

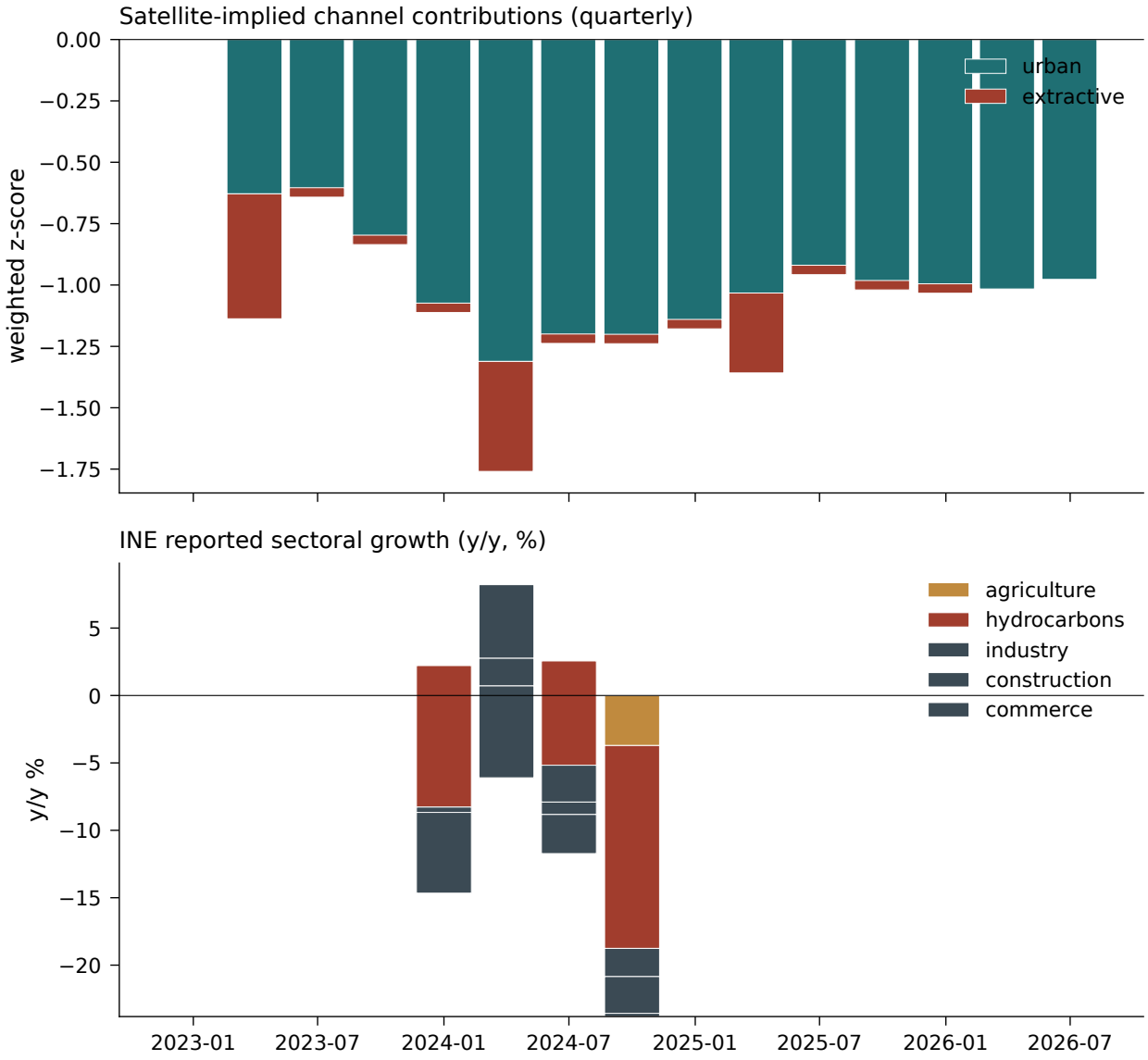

Figure 5: Sectoral decomposition of quarterly GDP growth. Top: implied from the satellite composite, with channels recovered from the two-factor DFM loadings. Bottom: INE's reported sectoral growth y/y. Disagreements between the two decompositions localize the recession sectorally.

6 Robustness

TBD: Robustness section to be written. Planned exercises:

- Alternative VIIRS products (`VNP46A3` vs. `DNB/MONTHLY_V1` vs. harmonized DMSP–VIIRS (Li et al., 2020)).
- Alternative flare-attribution radii (1, 2, 3 and 5 km) and temperature thresholds (1200, 1400 and 1600 K).

- Alternative TROPOMI products (OFFL vs. NRTI; `qa_value` cutoffs of 0.50, 0.75 and 0.90).
- Alternative DFM specifications (one, two, and three factors; AR(1) vs. AR(2) on factor and idiosyncratic).
- Alternative baseline periods for deseasonalization.
- Population-weighted vs. GDP-weighted composite across cities.
- Sub-period estimation: pre- vs. post-December-2025 fuel-subsidy elimination.

7 Conclusion

A four-stream satellite coincident indicator for Bolivia. The indicator covers approximately 78 percent of GDP at monthly frequency with a two-month publication lag, arriving two to three months before the quarterly national accounts and meeting the full twenty-four-vintage threshold of INE's experimental IGAE only in March 2028. **[TBD: Magnitude and timing of the satellite-implied recession vs. INE.]**

The remaining blind spot is non-tradeable services. Health, education, public administration, and domestic work together account for approximately 11 percent of GDP and are not visible to any of the four streams. A satellite composite that excludes them is a coincident indicator of *goods-producing and tradeable-services* activity, not of GDP in full. This caveat carries through every numerical comparison with INE; the magnitude of the satellite–INE gap is partly a feature of measurement coverage, not of manipulation alone.

Three directions for extension. (i) Extend the manipulation-detection framework to CPI using online-scraped prices in the Cavallo (2013) tradition. (ii) Pair the satellite composite with a high-frequency fiscal-dominance indicator combining BCB's CIN-SPNF with the parallel-rate premium. (iii) Apply the framework to the 2026 to 2027 Paz reform program in real time: the satellite indicator is well suited to testing whether stabilization is being captured in the national accounts or understated. The framework applies directly to Venezuela, Angola, and post-2021 Afghanistan.

The satellite data are free and the framework is open-source. Replication materials, processing pipeline, and live dashboard at `wernerhl.github.io/bolivia-ews`. The frozen replication dataset is archived at Zenodo DOI **[TBD: 10.5281/zenodo.XXXXXX]**.

A Spatial coverage and field-level calibration

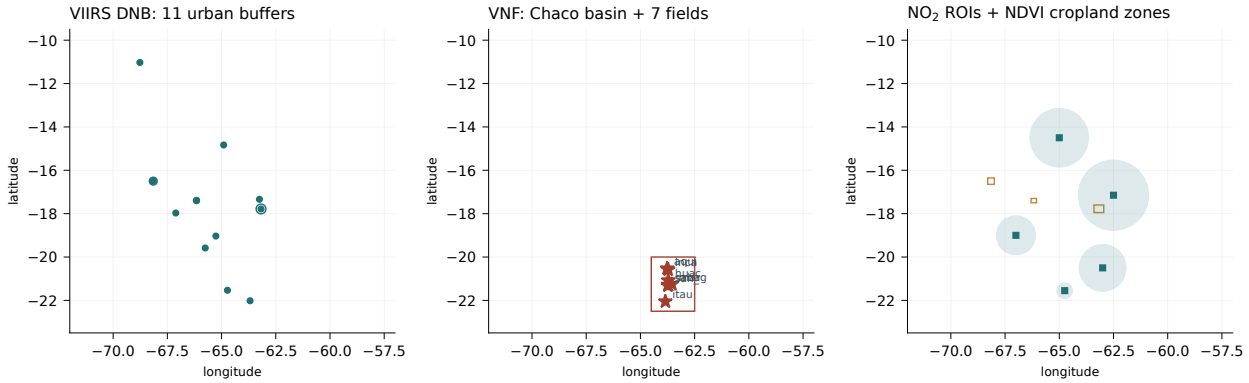

Figure 6: Spatial coverage of the four satellite streams. Left: urban nighttime-lights buffers. Center: Chaco bounding box and seven gas-field centroids. Right: TROPOMI NO_2 rectangles and Sentinel-2 cropland zones.

Figure 7: Field-level pre-crisis calibration of VIIRS Nightfire radiant heat against YPFB-reported gas production, 2012–2022. Slope, HAC standard error, and R^2 reported per field when data is available.

B Data vintages and robustness

Table 5: Data vintages, release lags, and revision behavior as of the submission vintage. Lag = nominal days between period end and source publication.

Series	Source	Freq.	Lag (d)	Coverage in paper	Revision
INE quarterly GDP	INE Nextcloud xlsx	Q	60–90	1990Q1–2024Q4	stable post-vintage
INE IGAE	INE portal	M	45	starts Mar 2026	preliminary
INE departmental GDP	INE annual xlsx	A	365–700	—	stable
YPFB gas prod. (total)	YPFB JPG chart	A	90	2006–2025	preliminary
YPFB field-month prod.	Boletín Estadístico	M	30–60	—	stable
IBCH cement dispatches	IBCH publications	M	20–40	—	stable
CNDC electricity	CNDC dload.php xlsx	M	5–10	—	stable
SIN tax collections	SIN portal	M	30	—	stable
Aduana imports	INE Nextcloud xlsx	M	30–45	2024-01–2026-02	stable
BCB monetary	BCB SPM	M	15–30	—	revised
VIIRS DNB (VNP46A2)	GEE	D	3–5	2012-01–2026-04	BRDF gap-filled
VIIRS Nightfire	EOG	D	1–2	—	stable
TROPOMI NO_2	GEE OFFL	D	5–10	2018-07–2026-04	QA pre-filtered
Sentinel-2 NDVI	GEE HARMONIZED	D	3–5	2017-03–(pend.)	SCL-masked
WB GGFR flaring	WB xlsx	A	180	2012–2024	stable
Binance P2P premium	Binance P2P API	D	0	2026-04-24+	real-time

Table 6: Robustness grid: trough date and depth under alternative specifications. Cells marked — indicate the specification has not yet been exercised in the current vintage; see notes below.

Specification	Trough date	Trough σ	2025 (%)	2026 (%)
Baseline (two-factor DFM, 2012-baseline)	2024-01	-2.62	—	—
VIIRS only	—	—	—	—
VNF only	—	—	—	—
NO_2 only	—	—	—	—
NDVI only	—	—	—	—
VNF radius 1 km / 2 km / 3 km	—	—	—	—
VNF threshold 1200 K / 1400 K / 1600 K	—	—	—	—
TROPOMI qa_value 0.50 / 0.75 / 0.90	—	—	—	—
One-factor vs two-factor vs weighted composite	—	—	—	—
Baseline 2013–2019 vs 2015–2019 vs 2017–2019	—	—	—	—
Population vs GDP weights	—	—	—	—

C Data processing details

[TBD: Appendix to be written. Planned contents: exact GEE code snippets for each stream, EOG authentication procedure, cloud-masking details, stray-light filter for TROPOMI, temperature-threshold sensitivity for VNF.]

D State-space estimation details

[TBD: Appendix to be written. Planned contents: full likelihood derivation, EM algorithm convergence criteria, Kalman filter initialization, treatment of ragged-edge release dates, news-decomposition formulas.]

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